Digital logic

National 5 · Electronics & control · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…
- ☐ complete truth tables for AND, OR and NOT and combinations of gates
- ☐ turn a written problem into a logic circuit

I DO — Worked example

Read how a full-mark answer is built

A machine guard system: the motor runs (Z) only when the guard is closed (A) AND the start button is pressed (B), OR when the maintenance key (C) is on.

1 Draw the logic diagram. (3 marks)

Points that earn marks from the marking instructions: one ✓ per point
- ✓ AND gate with inputs A and B
- ✓ OR gate with the AND output and C
- ✓ Output Z from the OR gate

WE DO — Guided practice

Use the starters and word bank to help you

A circuit has inputs A and B into an OR gate whose output goes through a NOT gate to Z.

2 Complete the truth table for Z. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) NOT A into an OR gate with B, then AND with C gives…

☐ $Z = \bar{A} + B \cdot C$ ☐ $Z = \bar{A} \cdot B + C$ ☐ $Z = (\bar{A} + B) \cdot C$ ☐ $Z = (A + B) \cdot C$

b) A=1, B=0 into an OR gate gives…

☐ 0 ☐ 2 ☐ Undefined ☐ 1

c) $Z = (A \cdot B) + C$. When A = 1, B = 0, C = 0, Z is…

☐ 2 ☐ 1 ☐ Undefined ☐ 0

d) A=1, B=0 into an AND gate gives…

☐ Both ☐ 1 ☐ 0 ☐ Neither

4 Key terms: write the letter of the matching description next to each term.

1. **Writing a Boolean expression**	____	**A** Output 1 only when all inputs are 1.
2. **AND gate**	____	**B** Output 1 when any input is 1.
3. **OR gate**	____	**C** Gate by gate from the inputs, with brackets for the order.

5 Explain what is meant by ‘NOT gate’. (2 marks)

6 Explain what is meant by ‘Rows in a 3-input truth table’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?